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Inventor(s)	Ekman; Matthew et al.

Auto-injector

Abstract

An auto-injector for administering a dose of a liquid medicament includes an elongate housing arranged to contain a syringe with a hollow needle and a stopper for sealing the syringe and displacing the medicament, the housing having a distal end and a proximal end with an orifice intended to be applied against an injection site. The syringe is slidably arranged with respect to the housing. A spring capable of, upon activation: pushing the needle from a covered position inside the housing into an advanced position through the orifice and past the proximal end (P), operating the syringe to supply the dose of medicament (M), and retracting the syringe with the needle into the covered position. After delivering the medicament, an activator arranged to lock the spring in a pressurized state prior to manual operation and capable of, upon manual operation, releasing the spring for injection.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3712301	12/1972	Sarnoff	N/A	N/A
4394863	12/1982	Bartner	N/A	N/A
4976701	12/1989	Ejlersen et al.	N/A	N/A
5137516	12/1991	Rand et al.	N/A	N/A
5267963	12/1992	Bachynsky	N/A	N/A
5300030	12/1993	Crossman et al.	N/A	N/A
5358489	12/1993	Wyrick	N/A	N/A
5643214	12/1996	Marshall et al.	N/A	N/A
5820602	12/1997	Kovelman et al.	N/A	N/A
6004297	12/1998	Steenfeldt-Jensen et al.	N/A	N/A
6277099	12/2000	Strowe et al.	N/A	N/A
6371939	12/2001	Bergens et al.	N/A	N/A
6454743	12/2001	Weber	N/A	N/A
6620137	12/2002	Kirchhofer et al.	N/A	N/A
7097634	12/2005	Gilbert	N/A	N/A
7195616	12/2006	Diller et al.	N/A	N/A
7291132	12/2006	DeRuntz et al.	N/A	N/A
7297135	12/2006	Jeffrey	N/A	N/A

7341575	12/2007	Rice et al.	N/A	N/A
7597685	12/2008	Olson	N/A	N/A
7666164	12/2009	Giambattista et al.	N/A	N/A
7678085	12/2009	Graf	N/A	N/A
7717877	12/2009	Lavi et al.	N/A	N/A
7771397	12/2009	Olson	N/A	N/A
7771398	12/2009	Knight et al.	N/A	N/A
7811261	12/2009	Rubinstein et al.	N/A	N/A
7901377	12/2010	Harrison	604/157	A61M 5/2033
7918824	12/2010	Bishop et al.	N/A	N/A
7976494	12/2010	Kohlbrenner et al.	N/A	N/A
7976499	12/2010	Grunhut et al.	N/A	N/A
7976514	12/2010	Abry	604/218	A61M 5/20
7988675	12/2010	Gillespie et al.	N/A	N/A
8038649	12/2010	Kronestedt	N/A	N/A
8048037	12/2010	Kohlbrenner et al.	N/A	N/A
8062255	12/2010	Brunnberg et al.	N/A	N/A
8083711	12/2010	Enggaard	N/A	N/A
8277412	12/2011	Kronestedt	N/A	N/A
8277414	12/2011	Barrow-Williams et al.	N/A	N/A
8323238	12/2011	Cronenberg et al.	N/A	N/A
8357125	12/2012	Grunhut et al.	N/A	N/A
8361025	12/2012	Lawlis et al.	N/A	N/A
8366680	12/2012	Raab	N/A	N/A
8376993	12/2012	Cox et al.	N/A	N/A
8376997	12/2012	Hogdahl et al.	N/A	N/A
8398594	12/2012	Streit et al.	N/A	N/A
8403883	12/2012	Fayyaz et al.	N/A	N/A
8409138	12/2012	James et al.	N/A	N/A
8409141	12/2012	Johnsen et al.	N/A	N/A
8409148	12/2012	Fiechter et al.	N/A	N/A
8414533	12/2012	Alexandersson	N/A	N/A
8439864	12/2012	Galbraith et al.	N/A	N/A
8491538	12/2012	Kohlbrenner et al.	N/A	N/A
8551054	12/2012	Guillermo	N/A	N/A
8568359	12/2012	Carrel et al.	N/A	N/A
8579865	12/2012	Wotton	604/218	A61K 9/0087
8579866	12/2012	Morgan et al.	N/A	N/A
8617109	12/2012	Kronestedt et al.	N/A	N/A
8617124	12/2012	Wieselblad et al.	N/A	N/A
8632507	12/2013	Bartha	N/A	N/A
8684969	12/2013	Moller et al.	N/A	N/A
8708973	12/2013	Holmqvist	N/A	N/A
8734394	12/2013	Adams et al.	N/A	N/A
8734402	12/2013	Sharp et al.	N/A	N/A
8758292	12/2013	Tschirren et al.	N/A	N/A
8808250	12/2013	Ekman et al.	N/A	N/A

8808251	12/2013	Raab et al.	N/A	N/A
8821451	12/2013	Daniel	N/A	N/A
8834431	12/2013	Kohlbrenner et al.	N/A	N/A
8840591	12/2013	Raab et al.	N/A	N/A
8882723	12/2013	Smith et al.	N/A	N/A
8911410	12/2013	Ekman et al.	N/A	N/A
8911411	12/2013	Nielsen	N/A	N/A
8939934	12/2014	Brereton et al.	N/A	N/A
8945063	12/2014	Wotton et al.	N/A	N/A
8956331	12/2014	Johnsen et al.	N/A	N/A
8961473	12/2014	Heald	N/A	N/A
8968256	12/2014	Raab	N/A	N/A
8968258	12/2014	Nzike et al.	N/A	N/A
8992484	12/2014	Radmer et al.	N/A	N/A
8992487	12/2014	Eich et al.	N/A	N/A
9005160	12/2014	Karlsson et al.	N/A	N/A
9011386	12/2014	Kronestedt et al.	N/A	N/A
9011387	12/2014	Ekman et al.	N/A	N/A
9022991	12/2014	Moeller	N/A	N/A
9022994	12/2014	Moser et al.	N/A	N/A
9044548	12/2014	Miller et al.	N/A	N/A
9044553	12/2014	James et al.	N/A	N/A
9057369	12/2014	Kohlbrenner et al.	N/A	N/A
9061104	12/2014	Daniel	N/A	N/A
9067024	12/2014	Roberts et al.	N/A	N/A
9089652	12/2014	Nzike et al.	N/A	N/A
9108002	12/2014	Markussen	N/A	N/A
9125988	12/2014	Karlsson	N/A	N/A
9132235	12/2014	Holmqvist	N/A	N/A
9199038	12/2014	Daniel	N/A	N/A
9205199	12/2014	Kemp et al.	N/A	N/A
9216256	12/2014	Olson et al.	N/A	N/A
9233213	12/2015	Olson et al.	N/A	N/A
9233214	12/2015	Kemp et al.	N/A	N/A
9233215	12/2015	Hourmand et al.	N/A	N/A
9242044	12/2015	Markussen	N/A	N/A
9272098	12/2015	Hourmand et al.	N/A	N/A
9283326	12/2015	Kemp et al.	N/A	N/A
9283327	12/2015	Hourmand et al.	N/A	N/A
9283328	12/2015	Dasbach	N/A	N/A
9308327	12/2015	Marshall et al.	N/A	N/A
9333304	12/2015	Brereton et al.	N/A	N/A
9339607	12/2015	Langley et al.	N/A	N/A
9352088	12/2015	Ekman et al.	N/A	N/A
9358345	12/2015	Brereton et al.	N/A	N/A
9358351	12/2015	Ekman et al.	N/A	N/A
9393368	12/2015	Nzike et al.	N/A	N/A
9402957	12/2015	Adams et al.	N/A	N/A
9408976	12/2015	Olson	N/A	N/A
9408977	12/2015	Butler et al.	N/A	N/A

9408979	12/2015	Veasey et al.	N/A	N/A
9415165	12/2015	Cowe	N/A	N/A
9421336	12/2015	Ekman et al.	N/A	N/A
9427525	12/2015	Barrow-Williams et al.	N/A	N/A
9446196	12/2015	Hourmand et al.	N/A	N/A
9446201	12/2015	Holmqvist	N/A	N/A
9457149	12/2015	Kemp et al.	N/A	N/A
9457152	12/2015	Raab et al.	N/A	N/A
9492622	12/2015	Brereton et al.	N/A	N/A
9649441	12/2016	Barrow-Williams et al.	N/A	N/A
9662452	12/2016	Daniel	N/A	N/A
9675758	12/2016	Johnston	N/A	N/A
9867940	12/2017	Holmqvist et al.	N/A	N/A
9872961	12/2017	Fourt et al.	N/A	N/A
10118001	12/2017	Fourt et al.	N/A	N/A
10314981	12/2018	Sampson et al.	N/A	N/A
10350356	12/2018	Hirschel et al.	N/A	N/A
10350362	12/2018	Dennis, Jr. et al.	N/A	N/A
10363377	12/2018	Atterbury et al.	N/A	N/A
10420898	12/2018	Daniel	N/A	N/A
10569019	12/2019	Hirschel et al.	N/A	N/A
10799647	12/2019	Hostettler et al.	N/A	N/A
10881799	12/2020	Hirschel et al.	N/A	N/A
11298462	12/2021	Atterbury et al.	N/A	N/A
11383044	12/2021	Tschirren et al.	N/A	N/A
11452821	12/2021	Lafever et al.	N/A	N/A
11484653	12/2021	Ekman et al.	N/A	N/A
11730888	12/2022	Ekman et al.	N/A	N/A
12083319	12/2023	Ekman et al.	N/A	N/A
12090302	12/2023	Ekman et al.	N/A	N/A
12090303	12/2023	Ekman et al.	N/A	N/A
2002/0007154	12/2001	Hansen et al.	N/A	N/A
2002/0095120	12/2001	Larsen et al.	N/A	N/A
2003/0105430	12/2002	Lavi et al.	N/A	N/A
2004/0024367	12/2003	Gilbert	N/A	N/A
2004/0039336	12/2003	Amark et al.	N/A	N/A
2005/0101919	12/2004	Brunnberg et al.	N/A	N/A
2005/0165360	12/2004	Stamp	N/A	N/A
2005/0203466	12/2004	Hommann et al.	N/A	N/A
2005/0222539	12/2004	Gonzales et al.	N/A	N/A
2006/0129089	12/2005	Stamp	N/A	N/A
2006/0153693	12/2005	Fiechter et al.	N/A	N/A
2006/0184117	12/2005	Knight et al.	N/A	N/A
2006/0287630	12/2005	Hommann	N/A	N/A
2007/0027430	12/2006	Hommann	N/A	N/A
2007/0112310	12/2006	Lavi et al.	N/A	N/A
2007/0135767	12/2006	Gillespie et al.	N/A	N/A

2007/0173770	12/2006	Stamp	604/187	A61M 5/3202
2008/0147006	12/2007	Brunnberg et al.	N/A	N/A
2008/0262427	12/2007	Hommann	N/A	N/A
2008/0269692	12/2007	James et al.	N/A	N/A
2009/0012471	12/2008	Harrison	N/A	N/A
2009/0054849	12/2008	Burnell et al.	N/A	N/A
2009/0264828	12/2008	Dette et al.	N/A	N/A
2010/0016793	12/2009	Jennings et al.	N/A	N/A
2010/0016795	12/2009	McLoughlin	N/A	N/A
2010/0049125	12/2009	James	604/110	A61M 5/2033
2010/0094214	12/2009	Abry	604/110	A61M 5/20
2010/0130930	12/2009	Stamp et al.	N/A	N/A
2010/0152655	12/2009	Stamp	N/A	N/A
2010/0185178	12/2009	Sharp et al.	N/A	N/A
2011/0054411	12/2010	Dowds et al.	N/A	N/A
2011/0178500	12/2010	Shang	604/506	A61P 29/00
2011/0282278	12/2010	Stamp et al.	N/A	N/A
2011/0319864	12/2010	Beller et al.	N/A	N/A
2012/0010575	12/2011	Jones et al.	N/A	N/A
2012/0041387	12/2011	Bruggemann et al.	N/A	N/A
2012/0053528	12/2011	Bollenbach et al.	N/A	N/A
2012/0172817	12/2011	Bruggemann et al.	N/A	N/A
2013/0035647	12/2012	Veasey et al.	N/A	N/A
2013/0041328	12/2012	Daniel	N/A	N/A
2013/0123710	12/2012	Ekman et al.	N/A	N/A
2013/0261556	12/2012	Jones et al.	N/A	N/A
2013/0274662	12/2012	Hourmand et al.	N/A	N/A
2013/0274677	12/2012	Ekman et al.	N/A	N/A
2013/0289525	12/2012	Kemp et al.	N/A	N/A
2013/0345643	12/2012	Hourmand et al.	N/A	N/A
2015/0273157	12/2014	Kohlbrenner et al.	N/A	N/A
2016/0089498	12/2015	Daniel	N/A	N/A
2018/0064875	12/2017	Holmqvist	N/A	N/A
2019/0374717	12/2018	Swanson et al.	N/A	N/A
2020/0139047	12/2019	Hirschel et al.	N/A	N/A
2021/0154407	12/2020	Hirschel et al.	N/A	N/A
2022/0211946	12/2021	Ekman et al.	N/A	N/A
2022/0323678	12/2021	Ekman et al.	N/A	N/A
2023/0022361	12/2022	Heiniger et al.	N/A	N/A
2023/0181827	12/2022	Ekman et al.	N/A	N/A
2024/0197997	12/2023	Ekman et al.	N/A	N/A
2024/0197998	12/2023	Ekman et al.	N/A	N/A
2024/0238520	12/2023	Ekman et al.	N/A	N/A
2024/0307621	12/2023	Ekman et al.	N/A	N/A
2024/0307622	12/2023	Ekman et al.	N/A	N/A
2024/0316269	12/2023	Ekman et al.	N/A	N/A
2024/0316270	12/2023	Ekman et al.	N/A	N/A
2024/0316271	12/2023	Ekman et al.	N/A	N/A

2024/0316272	12/2023	Ekman et al.	N/A	N/A
2024/0316273	12/2023	Ekman et al.	N/A	N/A
2024/0316274	12/2023	Ekman et al.	N/A	N/A
2024/0316275	12/2023	Ekman et al.	N/A	N/A
2025/0152819	12/2024	Ekman et al.	N/A	N/A
2025/0152820	12/2024	Ekman et al.	N/A	N/A
2025/0152821	12/2024	Ekman et al.	N/A	N/A
2025/0152823	12/2024	Ekman et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
705345	12/2012	CH	N/A
705992	12/2012	CH	N/A
19819409	12/1998	DE	N/A
102005038933	12/2006	DE	N/A
202007000578	12/2006	DE	N/A
102005052502	12/2006	DE	N/A
0666084	12/1994	EP	N/A
0824923	12/1997	EP	N/A
0991441	12/2002	EP	N/A
2399634	12/2010	EP	N/A
2468334	12/2011	EP	N/A
2468335	12/2011	EP	N/A
2742962	12/2013	EP	N/A
3650064	12/2019	EP	N/A
3381490	12/2019	EP	N/A
3184138	12/2022	EP	N/A
2654938	12/1990	FR	N/A
2438592	12/2006	GB	N/A
H08-502180	12/1995	JP	N/A
2002-528182	12/2001	JP	N/A
2007-500530	12/2006	JP	N/A
WO 1994/004103	12/1993	WO	N/A
WO 1995/035126	12/1994	WO	N/A
WO 1999/053979	12/1998	WO	N/A
WO 2000/024441	12/1999	WO	N/A
WO 2002/047746	12/2001	WO	N/A
WO 2003/062672	12/2002	WO	N/A
WO 2003/097133	12/2002	WO	N/A
WO 2005/009515	12/2004	WO	N/A
WO 2005/070481	12/2004	WO	N/A
WO 2005/097238	12/2004	WO	N/A
WO 2005/115507	12/2004	WO	N/A
WO 2006/057604	12/2005	WO	N/A
WO 2006/079064	12/2005	WO	N/A
WO 2007/083115	12/2006	WO	N/A
WO 2007/099044	12/2006	WO	N/A
WO 2007/129324	12/2006	WO	N/A
WO 2008/059385	12/2007	WO	N/A

WO 2008/116688	12/2007	WO	N/A
WO 2008/155377	12/2007	WO	N/A
WO 2009/019440	12/2008	WO	N/A
WO 2009/040601	12/2008	WO	N/A
WO 2009/040603	12/2008	WO	N/A
WO 2009/040607	12/2008	WO	N/A
WO 2009/040672	12/2008	WO	N/A
WO 2009/062508	12/2008	WO	N/A
WO 2009/081103	12/2008	WO	N/A
WO 2010/007395	12/2009	WO	N/A
WO 2010/063707	12/2009	WO	N/A
WO 2010/136077	12/2009	WO	N/A
WO 2011/012903	12/2010	WO	N/A
WO 2011/043714	12/2010	WO	N/A
WO 2011/075101	12/2010	WO	N/A
WO 2011/109205	12/2010	WO	N/A
WO 2011/111006	12/2010	WO	N/A
WO 2011/117592	12/2010	WO	N/A
WO 2011/123024	12/2010	WO	N/A
WO 2011/126439	12/2010	WO	N/A
WO 2012/045350	12/2011	WO	N/A
WO 2012/085024	12/2011	WO	N/A
WO 2019/074788	12/2018	WO	N/A
WO 2020/190529	12/2019	WO	N/A
WO 2021/008839	12/2020	WO	N/A
WO 2021/160540	12/2020	WO	N/A
WO 2021/197804	12/2020	WO	N/A
WO 2022/069617	12/2021	WO	N/A
WO 2022/184388	12/2021	WO	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability in Application No. PCT/EP2011/052301, dated Aug. 21, 2012, 6 pages. cited by applicant

International Search Report and Written Opinion in Application No. PCT/EP2011/052301, dated May 26, 2011, 9 pages. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a divisional of U.S. patent application Ser. No. 18/679,896, filed May 31, 2024, which is continuation of U.S. patent application Ser. No. 17/453,601, filed Nov. 4, 2021, which is a continuation of U.S. patent application Ser. No. 16/276,366, filed Feb. 14, 2019, now U.S. Patent Application No. 11, 197,958, which is a continuation of U.S. patent application Ser. No. 14/995,964, filed Jan. 14, 2016, now Patent No. 10,232, 116, which is a continuation of U.S. patent application Ser. No. 13/579,435, filed Feb. 25, 2013, now U.S. Pat. No. 9,248,245, which is a 35 U.S.C. 371 National Application of

PCT/EP2011/052301, filed Feb. 16, 2011, and claims priority to European Patent Application No. 10153999.7, filed Feb. 18, 2010 and U.S. Patent Application No. 61/412,086, filed Nov. 10, 2010, the entire contents of which are incorporated entirely herein by reference.

TECHNICAL FIELD

(1) The invention relates to an auto-injector for administering a dose of a liquid medicament according to the preamble of claim 1.

BACKGROUND OF THE INVENTION

(2) Administering an injection is a process which presents a number of risks and challenges for users and healthcare professionals, both mental and physical.

(3) Injection devices (i.e. devices capable of delivering medicaments from a medication container) typically fall into two categories—manual devices and auto-injectors.

(4) In a manual device—the user must provide the mechanical energy to drive the fluid through the needle. This is typically done by some form of button/plunger that has to be continuously pressed by the user during the injection. There are numerous disadvantages to the user from this approach. If the user stops pressing the button/plunger then the injection will also stop. This means that the user can deliver an underdose if the device is not used properly (i.e. the plunger is not fully pressed to its end position). Injection forces may be too high for the user, in particular if the patient is elderly or has dexterity problems.

(5) The extension of the button/plunger may be too great. Thus it can be inconvenient for the user to reach a fully extended button. The combination of injection force and button extension can cause trembling/shaking of the hand which in turn increases discomfort as the inserted needle moves.

(6) Auto-injector devices aim to make self-administration of injected therapies easier for patients. Current therapies delivered by means of self-administered injections include drugs for diabetes (both insulin and newer GLP-1 class drugs), migraine, hormone therapies, anticoagulants etc.

(7) Auto-injectors are devices which completely or partially replace activities involved in parenteral drug delivery from standard syringes. These activities may include removal of a protective syringe cap, insertion of a needle into a patient's skin, injection of the medicament, removal of the needle, shielding of the needle and preventing reuse of the device. This overcomes many of the disadvantages of manual devices. Injection forces/button extension, hand-shaking and the likelihood of delivering an incomplete dose are reduced. Triggering may be performed by numerous means, for example a trigger button or the action of the needle reaching its injection depth. In some devices the energy to deliver the fluid is provided by a spring.

(8) US 2002/0095120 A1 discloses an automatic injection device which automatically injects a pre-measured quantity of fluid medicine when a tension spring is released. The tension spring moves an ampoule and the injection needle from a storage position to a deployed position when it is released. The content of the ampoule is thereafter expelled by the tension spring forcing a piston forward inside the ampoule. After the fluid medicine has been injected, torsion stored in the tension spring is released and the injection needle is automatically retracted back to its original storage position.

(9) US 2007/0112310 A1 discloses an injector being automatic in that the needle is inserted into the injection site (e.g., a patient's skin) with user or caregiver assistance, the delivery is automatically initiated upon needle insertion, and the needle is retracted automatically after the end of delivery. Preferably the needle is not seen by the user prior to, during or after injection. Prior to and after injection, the needle is hidden in the device so as to avoid any potential injury or health risk to the user or health care provider. The injector includes a housing and a shield arranged to slide relative to the housing and a driver moving during drug delivery. The housing and shield form a cartridge enclosure. The cartridge is shielded and locked after delivery is completed. A needle-locking mechanism can be used in any number of pen-like injectors or safety needles.

(10) U.S. Pat. No. 5,267,963 discloses an automatic injection device which, upon activation by the user, automatically extends a syringe with needle, delivers medication through the needle, and

retracts the needle, thus keeping the needle hidden from view. All motions are achieved by means of a tension spring and a cam profile.

(11) WO 2009/081103 A1 discloses an auto-injector for a syringe that is suitable for use in the injected delivery of drug to a patient. The auto-injector comprises a housing defining a housing cavity arranged for receipt of a syringe; and a needle delivery aperture through which a needle tip of the syringe protrudes during dispensing of the liquid drug formulation. The auto-injector further comprises a barrel coupling element that couples to the barrel of the syringe and is movable in tandem therewith; a syringe advancer for moving the syringe and barrel coupling element in tandem therewith from a rest position, in which the hollow needle, is within the housing to a use position, in which the needle tip protrudes through the needle delivery aperture; a syringe actuator for actuating the syringe by plunging said plunger within the barrel of the syringe to dispense the liquid drug formulation; a syringe retractor for retracting the syringe and barrel coupling element in tandem therewith from the use position to a retract position, in which the hollow needle is within the housing; and a syringe lock for locking the barrel coupling element and syringe coupled thereto in the retract position.

(12) WO 2008/155377 discloses an inserter for an infusion set comprising an insertion needle and a spring unit assuring automatic insertion and automatic retraction of the insertion needle. The inserter for a medical device comprises—a housing, —a first body which is movable relative to the housing and comprising penetrating means pointing in the direction of insertion, —a second body which is also movable relative to the housing and—driving means which move respectively the first body and the second body relative to the housing wherein the driving means moves the first body in the direction of insertion and moves the second body in a direction different from the insertion direction.

(13) DE 10 2005 038 933 A1 discloses an inserter for an infusion set comprising an insertion needle and a spring unit assuring automatic insertion and automatic retraction of the insertion needle. The inserter for a medical device comprises—a housing, —a first body which is movable relative to the housing and comprising penetrating means pointing in the direction of insertion, —a second body which is also movable relative to the housing and—driving means which move respectively the first body and the second body relative to the housing wherein the driving means moves the first body in the direction of insertion and moves the second body in a direction different from the insertion direction.

SUMMARY OF THE INVENTION

(14) It is an object of the present invention to provide an improved auto-injector.

(15) The object is achieved by an auto-injector according to claim 1.

(16) Preferred embodiments of the invention are given in the dependent claims.

(17) According to the invention, an auto-injector for administering a dose of a liquid medicament comprises: an elongate housing arranged to contain a syringe with a hollow needle and a stopper for sealing the syringe and displacing the medicament, the housing having a distal end and a proximal end with an orifice intended to be applied against an injection site, wherein the syringe is slidably arranged with respect to the housing, spring means capable of, upon activation: pushing the needle from a covered position inside the housing into an advanced position through the orifice and past the proximal end, operating the syringe to supply the dose of medicament, and retracting the syringe with the needle into the covered position after delivering the medicament, activating means arranged to lock the spring means in a pressurized state prior to manual operation and capable of, upon manual operation, releasing the spring means for injection.

(18) In the context of this patent application the term proximal refers to the direction pointing towards the patient during an injection while the term distal refers to the opposite direction pointing away from the patient.

(19) According to the invention the spring means is a single compression spring arranged to be grounded at a distal end in the housing for advancing the needle and for injecting the dose of

medicament. The force of the compression spring is forwarded to the needle and/or the syringe via a plunger. The compression spring is arranged to have its ground in the housing switched to its proximal end for retracting the syringe when the injection of the medicament is at least nearly finished.

(20) The single compression spring is used for inserting the needle, fully emptying the syringe and retracting the syringe and needle to a safe position after injection. Thus a second spring for withdrawing the syringe and needle, which is a motion with an opposite sense compared to advancing the syringe and injecting the dose, is not required. While the distal end of the compression spring is grounded the proximal end moves the syringe forward for inserting the needle and carries on to the injection by pushing on the stopper. When the injection is at least nearly finished the compression spring bottoms out at its proximal end, resulting in the proximal end being grounded in the housing. At the same time the distal end of the compression spring is released from its ground in the housing. The compression spring is now pulling the syringe in the opposite direction.

(21) The auto-injector according to the invention has a particularly low part count compared to most conventional auto-injectors. The use of just one compression spring reduces the amount of metal needed and thus consequently reduces weight and manufacturing costs.

(22) In a preferred embodiment of the invention a retraction sleeve is axially movable arranged in the housing. At least one latch is provided for axially fixing the retraction sleeve in a maximum proximal position. The compression spring is arranged inside the retraction sleeve with its distal end bearing against a distal end face of the retraction sleeve and with its proximal end bearing against a thrust face of a decoupling member. The decoupling member is arranged to decouple the latch when being moved in proximal direction nearly into a maximum proximal position. When decoupled the retraction sleeve is allowed to move in distal direction and retract the needle by means of the spring force which is no longer grounded at its distal end.

(23) Preferably the plunger is arranged for pushing the syringe and/or the stopper in proximal direction. At least two resilient decoupling arms are arranged at the decoupling member. The decoupling arms exhibit inner ramped surfaces bearing against a first shoulder of the plunger in proximal direction P. The resilient decoupling arms are supportable by an inner wall of the retraction sleeve in order to prevent the decoupling arms from being flexed outward and slip past the first shoulder. In this state the plunger may be pushed in proximal direction by the decoupling member pushing against the first shoulder in order to insert the needle and inject the dose. At least one aperture is arranged in the retraction sleeve allowing the decoupling arms to be flexed outward by the first shoulder thus allowing the first shoulder to slip through the decoupling arms in proximal direction. This may happen when the injection is at least nearly finished. The decoupled plunger allows the syringe and needle to be retracted since it is no longer bearing against the decoupling member.

(24) The syringe may be arranged for joint axial movement with a syringe holder which is slidably arranged in the retraction sleeve. The syringe holder is provided with at least two resilient syringe holder arms arranged distally, the syringe holder arms having a respective inclined surface for bearing against a second shoulder, which is arranged at the plunger proximally from the first shoulder. The syringe holder arms are supportable by an inner surface of the housing in order to prevent them from being flexed outward. Thus, when the trigger button is pressed the spring force forwarded by the plunger does not yet press against the stopper but against the syringe for forwarding it. Consequently, a so called wet injection is avoided, i.e. the liquid medicament is not leaking out of the hollow needle before the needle is inserted. A widened portion is provided in the housing for allowing the syringe holder arms to flex outwards when the syringe holder has nearly reached a maximum proximal position thus allowing the second shoulder to slip through the syringe holder arms and to switch load of the compression spring from the syringe to the stopper. This allows for defining the moment to start injecting the medicament.

(25) A stud may be arranged at the distal end of the plunger. The retraction sleeve may have two or more resilient arms distally from the end face for holding the stud. The stud and/or the resilient arms have ramp features. Thus the resilient arms may be pushed apart by the stud when the plunger is moved in proximal direction. The activating means comprise a trigger button arranged at the distal end of the auto-injector. The trigger button is axially moveable and has at least two rigid retainers for preventing the resilient arms from being flexed outward when the trigger button is in a maximum distal position. Upon pushing the trigger button in proximal direction the retainers are moved in proximal direction in a manner to allow the resilient arms to be flexed out by the stud biased by the compression spring in proximal direction. Thus the stud is allowed to slip past the resilient arms in proximal direction under load of the compression spring in order to start a needle insertion/injection/retraction cycle. The main advantages of this trigger mechanism are its simplicity, the low part count and a high reliability.

(26) In order to reduce the risk of unintentionally triggering the auto-injector a safety button may be arranged laterally at the housing. The safety button has an interlock for preventing the trigger button from being pushed. The safety button is arranged to pull the interlock outward when operated thus allowing the trigger button to be pushed. For this purpose the safety button may be pivoted in the housing or it may be cast in one piece with the housing in a manner to be pivoted somewhere in the middle so pushing one end inwards causes the other end to be pulled outwards.

(27) Consequently, in order to operate the trigger button the safety button has to be pushed first so the auto-injector cannot be operated unintentionally. Another advantage of the lateral safety button is that the risk of operating the auto-injector in the wrong orientation and injecting into the thumb is reduced.

(28) In a preferred embodiment of the invention a delay box is arranged for slowing down the motion of the retraction sleeve. The latches are arranged to be disengaged by the decoupling member before the stopper has reached a maximum proximal position in the syringe. The apertures are arranged to meet the decoupling arms after the stopper has reached its maximum proximal position by means of the motion of the retraction sleeve. A gap is provided between a front face of the retraction sleeve and the syringe holder in their respective maximum proximal positions. The gap allows the retraction sleeve to travel a distance before retracting the syringe holder so the syringe holder is retracted after the decoupling arms met the apertures.

(29) Triggering the retraction when the stopper exactly reaches the end of its travel is a problem due to tolerances when manufacturing the syringe and stopper. Due to these tolerances the position of the stopper at the end of its travel is not repeatable. Consequently, in some cases the stopper would prematurely bottom out so the retraction would not be triggered at all. In other cases the retraction would be triggered before the stopper bottomed so residual medicament would remain in the syringe.

(30) Releasing the retraction sleeve from the housing a certain amount of time or travel before the stopper bottoms out in the syringe avoids the risk of stalling the retraction by the stopper hitting the end of the syringe prematurely. The damped backward motion of the retraction sleeve due to the delay box allows the plunger and stopper to finish their forward travel so the syringe is entirely emptied. The apertures of the retraction sleeve and the decoupling arms, which are now moving in opposite directions, meet after the stopper and plunger have stopped in order to decouple the decoupling member from the plunger. Due to the gap between the front face and the syringe holder the retraction sleeve is not immediately dragging the syringe back in distal direction when starting to move back. When the retraction sleeve has travelled back far enough to close the gap the stopper has already bottomed out and the plunger has been decoupled from the decoupling member. As soon as the gap is closed the syringe holder, the syringe, the hollow needle and the plunger are dragged back in distal direction.

(31) Thus both problems are solved, reliably retracting the hollow needle to a safe position and fully emptying the syringe which is particularly desirable with expensive drugs. Emptying the

syringe is also important for dosage accuracy.

(32) The delay box may comprise a circumferential outer wall with a back collar attached to the housing and a circumferential inner wall with a front collar attached to the retraction sleeve. A volume is defined between the outer wall and inner wall, the volume sealed by the back collar and front collar and filled with a viscous fluid. At least one hole is arranged in the delay box for allowing the viscous fluid to be pushed out as the volume decreases due to motion of the retraction sleeve. This is a particularly simple and cost-efficient way to damp the backward motion of the retraction sleeve.

(33) Usually the hollow needle is equipped with a protective needle shield for keeping the needle sterile and preventing it from being mechanically damaged. The protective needle shield is attached to the needle when the auto-injector or the syringe is assembled.

(34) Preferably a cap is provided at the proximal end of the housing. A sheet metal clip is attached to the cap for joint axial movement and independent rotation. The sheet metal clip is arranged to extend through an orifice into the housing when the cap is attached to the housing. The sheet metal clip comprises at least two barbs snapped into a circumferential notch or behind a shoulder of the protective needle shield. This allows for automatically engaging the sheet metal clip with the protective needle shield during assembly. When the cap is removed from the housing in preparation of an injection the protective needle shield is reliably removed without exposing the user too high a risk to injure themselves.

(35) The cap may be attachable to the housing by a screw connection. This allows for a low force removal of the protective needle shield.

(36) The housing may have at least one viewing window for inspecting the syringe.

(37) The auto-injector may preferably be used for subcutaneous or intra-muscular injection, particularly for delivering one of an analgetic, an anticoagulant, insulin, an insulin derivate, heparin, Lovenox, a vaccine, a growth hormone, a peptide hormone, a proteine, antibodies and complex carbohydrates.

(38) The term “medicament”, as used herein, means a pharmaceutical formulation containing at least one pharmaceutically active compound,

(39) wherein in one embodiment the pharmaceutically active compound has a molecular weight up to 1500 Da and/or is a peptide, a proteine, a polysaccharide, a vaccine, a DNA, a RNA, a antibody, an enzyme, an antibody, a hormone or an oligonucleotide, or a mixture of the above-mentioned pharmaceutically active compound, wherein in a further embodiment the pharmaceutically active compound is useful for the treatment and/or prophylaxis of diabetes mellitus or complications associated with diabetes mellitus such as diabetic retinopathy, thromboembolism disorders such as deep vein or pulmonary thromboembolism, acute coronary syndrome (ACS), angina, myocardial infarction, cancer, macular degeneration, inflammation, hay fever, atherosclerosis and/or rheumatoid arthritis, wherein in a further embodiment the pharmaceutically active compound comprises at least one peptide for the treatment and/or prophylaxis of diabetes mellitus or complications associated with diabetes mellitus such as diabetic retinopathy,

(40) wherein in a further embodiment the pharmaceutically active compound comprises at least one human insulin or a human insulin analogue or derivative, glucagon-like peptide (GLP-1) or an analogue or derivative thereof, or exedin-3 or exedin-4 or an analogue or derivative of exedin-3 or exedin-4.

(41) Insulin analogues are for example Gly (A21), Arg (B31), Arg (B32) human insulin; Lys (B3), Glu (B29) human insulin; Lys (B28), Pro (B29) human insulin; Asp (B28) human insulin; human insulin, wherein proline in position B28 is replaced by Asp, Lys, Leu, Val or Ala and wherein in position B29 Lys may be replaced by Pro; Ala (B26) human insulin; Des (B28-B30) human insulin; Des (B27) human insulin and Des (B30) human insulin.

(42) Insulin derivates are for example B29-N-myristoyl-des (B30) human insulin; B29-N-palmitoyl-des (B30) human insulin; B29-N-myristoyl human insulin; B29-N-palmitoyl human

insulin; B28-N-myristoyl-LysB28ProB29 human insulin; B28-N-palmitoyl-LysB28ProB29 human insulin; B30-N-myristoyl-ThrB29LysB30 human insulin; B30-N-palmitoyl-ThrB29LysB30 human insulin; B29-N-(N-palmitoyl-Y-glutamyl)-des (B30) human insulin; B29-N-(N-lithocholyl-Y-glutamyl)-des (B30) human insulin; B29-N-(ω -carboxyheptadecanoyl)-des (B30) human insulin and B29-N-(ω -carboxyheptadecanoyl) human insulin.

(43) Exendin-4 for example means Exendin-4 (1-39), a peptide of the sequence H-His-Gly-Glu-Gly-Thr-Phe-Thr-Ser-Asp-Leu-Ser-Lys-Gln-Met-Glu-Glu-Glu-Ala-Val-Arg-Leu-Phe-11e-Glu-Trp-Leu-Lys-Asn-Gly-Gly-Pro-Ser-Ser-Gly-Ala-Pro-Pro-Ser-NH₂.

(44) Exendin-4 derivatives are for example selected from the following list of compounds: H-(Lys) 4-des Pro36, des Pro37 Exendin-4 (1-39)-NH₂, H-(Lys) 5-des Pro36, des Pro37 Exendin-4 (1-39)-NH₂, des Pro36 [Asp28] Exendin-4 (1-39), des Pro36 [IsoAsp28] Exendin-4 (1-39), des Pro36 [Met(O) 14, Asp28] Exendin-4 (1-39), des Pro36 [Met(O) 14, IsoAsp28] Exendin-4 (1-39), des Pro36 [Trp(O2) 25, Asp28] Exendin-4 (1-39), des Pro36 [Trp(O2) 25, IsoAsp28] Exendin-4 (1-39), des Pro36 [Met(O) 14 Trp(O2) 25, Asp28] Exendin-4 (1-39), des Pro36 [Met(O) 14 Trp(O2) 25, IsoAsp28] Exendin-4 (1-39); or des Pro36 [Asp28] Exendin-4 (1-39), des Pro36 [IsoAsp28] Exendin-4 (1-39), des Pro36 [Met(O) 14, Asp28] Exendin-4 (1-39), des Pro36 [Met(O) 14, IsoAsp28] Exendin-4 (1-39), des Pro36 [Trp(O2) 25, Asp28] Exendin-4 (1-39), des Pro36 [Trp(O2) 25, IsoAsp28] Exendin-4 (1-39), des Pro36 [Met(O) 14 Trp(O2) 25, Asp28] Exendin-4 (1-39), des Pro36 [Met(O) 14 Trp(O2) 25, IsoAsp28] Exendin-4 (1-39), wherein the group-Lys6-NH₂ may be bound to the C-terminus of the Exendin-4 derivative; or an Exendin-4 derivative of the sequence H-(Lys) 6-des Pro36 [Asp28] Exendin-4 (1-39)-Lys6-NH₂, des Asp28 Pro36, Pro37, Pro38Exendin-4 (1-39)-NH₂, H-(Lys) 6-des Pro36, Pro38 [Asp28] Exendin-4 (1-39)-NH₂, H-Asn-(Glu) 5des Pro36, Pro37, Pro38 [Asp28] Exendin-4 (1-39)-NH₂, des Pro36, Pro37, Pro38 [Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂, H-(Lys) 6-des Pro36, Pro37, Pro38 [Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂, H-Asn-(Glu) 5-des Pro36, Pro37, Pro38 [Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂, H-(Lys) 6-des Pro36 [Trp(O2) 25, Asp28] Exendin-4 (1-39)-Lys6-NH₂, H-des Asp28 Pro36, Pro37, Pro38 [Trp(O2) 25] Exendin-4 (1-39)-NH₂, H-(Lys) 6-des Pro36, Pro37, Pro38 [Trp(O2) 25, Asp28] Exendin-4 (1-39)-NH₂, H-Asn-(Glu) 5-des Pro36, Pro37, Pro38 [Trp(O2) 25, Asp28] Exendin-4 (1-39)-NH₂, des Pro36, Pro37, Pro38 [Trp(O2) 25, Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂, H-(Lys) 6-des Pro36, Pro37, Pro38 [Trp(O2) 25, Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂, H-Asn-(Glu) 5-des Pro36, Pro37, Pro38 [Trp(O2) 25, Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂, H-(Lys) 6-des Pro36 [Met(O) 14, Asp28] Exendin-4 (1-39)-Lys6-NH₂, des Met(O) 14 Asp28 Pro36, Pro37, Pro38 Exendin-4 (1-39)-NH₂, H-(Lys) 6-desPro36, Pro37, Pro38 [Met(O) 14, Asp28] Exendin-4 (1-39)-NH₂, H-Asn-(Glu) 5-des Pro36, Pro37, Pro38 [Met(O) 14, Asp28] Exendin-4 (1-39)-NH₂, des Pro36, Pro37, Pro38 [Met(O) 14, Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂, H-(Lys) 6-des Pro36, Pro37, Pro38 [Met(O) 14, Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂, H-Asn-(Glu) 5 des Pro36, Pro37, Pro38 [Met(O) 14, Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂, H-Lys6-des Pro36 [Met(O) 14, Trp(O2) 25, Asp28] Exendin-4 (1-39)-Lys6-NH₂, H-des Asp28 Pro36, Pro37, Pro38 [Met(O) 14, Trp(O2) 25] Exendin-4 (1-39)-NH₂, H-(Lys) 6-des Pro36, Pro37, Pro38 [Met(O) 14, Asp28] Exendin-4 (1-39)-NH₂, H-Asn-(Glu) 5-des Pro36, Pro37, Pro38 [Met(O) 14, Trp(O2) 25, Asp28] Exendin-4 (1-39)-NH₂, des Pro36, Pro37, Pro38 [Met(O) 14, Trp(O2) 25, Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂, H-(Lys) 6-des Pro36, Pro37, Pro38 [Met(O) 14, Trp(O2) 25, Asp28] Exendin-4 (S1-39)-(Lys) 6-NH₂, H-Asn-(Glu) 5-des Pro36, Pro37, Pro38 [Met(O) 14, Trp(O2) 25, Asp28] Exendin-4 (1-39)-(Lys) 6-NH₂; or a pharmaceutically acceptable salt or solvate of any one of the aforementioned Exedin-4 derivative.

(45) Hormones are for example hypophysis hormones or hypothalamus hormones or regulatory active peptides and their antagonists as listed in Rote Liste, ed. 2008, Chapter 50, such as Gonadotropine (Follitropin, Lutropin, Choriongonadotropin, Menotropin), Somatropine (Somatropin), Desmopressin, Terlipressin, Gonadorelin, Triptorelin, Leuprorelin, Buserelin, Nafarelin, Goserelin.

(46) A polysaccharide is for example a glucosaminoglycane, a hyaluronic acid, a heparin, a low molecular weight heparin or an ultra low molecular weight heparin or a derivative thereof, or a sulphated, e.g. a poly-sulphated form of the above-mentioned polysaccharides, and/or a pharmaceutically acceptable salt thereof. An example of a pharmaceutically acceptable salt of a poly-sulphated low molecular weight heparin is enoxaparin sodium.

(47) Pharmaceutically acceptable salts are for example acid addition salts and basic salts. Acid addition salts are e.g. HCl or HBr salts. Basic salts are e.g. salts having a cation selected from alkali or alkaline, e.g. Na.sup.+, or K.sup.+, or Ca.sup.2+, or an ammonium ion N+(R1)(R2)(R3)(R4), wherein R1 to R4 independently of each other mean: hydrogen, an optionally substituted C1-C6-alkyl group, an optionally substituted C2-C6-alkenyl group, an optionally substituted C6-C10-aryl group, or an optionally substituted C6-C10-heteroaryl group. Further examples of pharmaceutically acceptable salts are described in "Remington's Pharmaceutical Sciences" 17. ed. Alfonso R. Gennaro (Ed.), Mark Publishing Company, Easton, Pa., U.S.A., 1985 and in Encyclopedia of Pharmaceutical Technology.

(48) Pharmaceutically acceptable solvates are for example hydrates.

(49) The delay box may be employed with other types of auto-injectors.

(50) The cap with the sheet metal spring may also be applied with other auto-injectors and injection devices.

(51) Further scope of applicability of the present invention will become apparent from the detailed description given hereinafter. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the invention, are given by way of illustration only, since various changes and modifications within the spirit and scope of the invention will become apparent to those skilled in the art from this detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will become more fully understood from the detailed description given herein below and the accompanying drawings which are given by way of illustration only, and thus, are not limiting of the present invention, and wherein:

(2) FIG. 1 are two longitudinal sections of an auto-injector with a single compression spring for advancing a syringe with a needle, injecting a dose of medicament and retracting the syringe and needle, the auto-injector as-delivered,

(3) FIG. 2 are two longitudinal sections of the auto-injector with the syringe and needle advanced and the dose expelled from the syringe,

(4) FIG. 3 is a perspective sectional view of the auto-injector in the initial state of FIG. 1,

(5) FIG. 4 is another perspective sectional view of the auto-injector of FIG. 3, and

(6) FIG. 5 is a detail view of the distal end of the auto-injector with a delay box,

(7) FIG. 6 is a detailed view of the proximal end of the autoinjector showing the cap and needle shield remover.

(8) Corresponding parts are marked with the same reference symbols in all figures.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(9) FIG. 1 shows two longitudinal sections in different section planes of an auto-injector 1, the different section planes approximately 90° rotated to each other. The auto-injector 1 comprises an elongate housing 2. A syringe 3, e.g. a Hypak syringe, with a hollow needle 4 is arranged in a proximal part of the auto-injector 1. When the auto-injector 1 or the syringe 3 is assembled a protective needle shield 5 is attached to the needle 4. A stopper 6 is arranged for sealing the syringe 3 distally and for displacing a liquid medicament M through the hollow needle 4. The syringe 3 is held in a tubular syringe carrier 7 and supported at its proximal end therein. A single compression

spring **8** is arranged in a distal part of the auto-injector **1**. A plunger **9** is arranged for forwarding the spring force of the compression spring **8**.

(10) Inside the housing **2** a retraction sleeve **10** is slidably arranged. Before the injection is triggered as shown in FIG. **1** the retraction sleeve **10** is in a maximum proximal position and prevented from moving in distal direction **D** by means of stops **11** caught behind latches **12** in the housing **2**. A distal end of the compression spring **8** bears against an end face **13** of the retraction sleeve **10**. Due to the stops **11** and latches **12** the force of the compression spring **8** is thus reacted into the housing **2**. The proximal end of the compression spring **8** bears against a decoupling member **14** arranged around the plunger **9**. Distally from the end face **13** the retraction sleeve has two or more resilient arms **15** for holding a stud **16** and keeping it from being moved in proximal direction **P**. The stud **16** is arranged at the distal end of the plunger **9**. The stud **16** and the resilient arms **15** have corresponding ramp features for pushing the resilient arms **15** apart in order to allow the stud **16** and the plunger **9** to move in proximal direction **P**.

(11) The decoupling member **14** comprises a thrust face **17** for bearing against a proximal end of the compression spring **8**. Proximally from the thrust face **17** two or more resilient decoupling arms **18** are provided at the decoupling member **14**, the decoupling arms **18** having inner ramped surfaces bearing against a first shoulder **19** in the plunger **9** in proximal direction **P**. The resilient decoupling arms **18** are supported by an inner wall of the retraction sleeve **10** in this situation so they cannot flex outward and slip past the first shoulder **19**.

(12) A trigger button **20** is arranged at the distal end **D** of the auto-injector **1**. The trigger button **20** may be pushed in proximal direction **P** in order to start an injection. As long as the trigger button **20** is not pushed the resilient arms **15** are caught between two or more retainers **21** arranged at the trigger button **20** so the resilient arms **15** cannot flex outward and the stud **16** although proximally biased by the compression spring **8** cannot slip through.

(13) The syringe carrier **7** is engaged for joint axial movement with a syringe holder **22** which is slidably arranged in the retraction sleeve **10**. The syringe holder **22** is provided with two or more resilient syringe holder arms **23** arranged distally. The syringe holder arms **23** have a respective inclined surface for bearing against a second shoulder **24** in the plunger **9** arranged proximally from the first shoulder **19**. In the initial position shown in FIG. **1** the syringe holder arms **23** are supported by an inner surface of the housing **2** so they cannot flex outward and the second shoulder **24** cannot slip through. In order to support the syringe holder arms **23** at the housing **2** a respective number of apertures are provided in the retraction sleeve **10**.

(14) FIG. **1** shows the auto-injector **1** as-delivered with a cap **25** screwed onto to the proximal end **P** of the auto-injector **1**. FIG. **6** shows details of the proximal end **P** with the cap **25**. The cap **25** comprises a sheet metal clip **26** with two or more barbs **27** extending through an orifice into the proximal end **P** of the auto-injector **1**. The sheet metal clip **26** is mounted to the cap **25** for joint axial movement with respect to a **(20)** longitudinal axis of the auto-injector **1**. However, the sheet metal clip **26** may rotate independently from the cap **25**. This may be achieved by attaching the sheet metal clip **26** with a hole in its base onto a pin protruding inwardly from the cap **25** and deforming the pin to form a mushroom-shaped closing head **28** so as to prevent the sheet metal clip **26** from being removed while allowing some clearance for the sheet metal clip **26** to rotate. When the cap **25** is screwed onto the proximal **P** end of the auto-injector **1** the barbs **27** are pushed down the protective needle shield **5** and snap into a circumferential notch arranged in the protective needle shield **5** or behind a shoulder thereof.

(15) When a user wants to operate the auto-injector **1** the first step is to unscrew the cap **25**. Thus the barbs **27** pull the protective needle shield **5** off the syringe **3** in proximal direction **P** and through the orifice making the syringe **3** ready to be used.

(16) A safety button **29** is arranged laterally at the distal part of the housing **2**. The safety button **29** serves for interlocking with the trigger button **20** in a manner to prevent the trigger button **20** from being inadvertently operated without the safety button **29** being released from a first blocking

position.

(17) Consequently, in order to operate the trigger button **20** the safety button **29** has to be pushed transversally with respect to the longitudinal axis against the force of a spring element **30** which is formed in the safety button **29**. The safety button **29** is pivoted in the middle so pushing the proximal end of the safety button **29** inward pulls an interlock **31** at its proximal end obstructing the trigger button **20** outward so the trigger button **20** can be pushed.

(18) When the trigger button **20** is pushed the retainers **21** are pushed in proximal direction P so the resilient arms **15** are allowed to flex outward. Under load of the compression spring **8** the inclined surfaces of the stud **16** force the resilient arms **15** apart until the stud **16** can slip through.

(19) The second shoulder **24** pushes the syringe holder **22**, syringe carrier **7** and syringe **3** forward while no load is exerted onto the stopper **6**. The hollow needle **4** appears from the proximal end P and is inserted into an injection site, e.g. a patient's skin.

(20) The forward movement continues until the syringe holder **22** bottoms out at a first abutment **32** in the housing **2** (see FIG. 2). The travel from the initial position (cf. FIG. 1) up to this point defines an injection depth, i.e. needle insertion depth.

(21) When the syringe holder **22** has nearly bottomed out the resilient syringe holder arms **23** have reached a widened portion **2.1** of the housing **2** where they are no longer supported by the inner wall of the housing **2**. However, since the force required to insert the needle **4** is relatively low the second shoulder **24** will continue to drive forward the syringe holder **22** until proximal travel is halted at the first abutment **32**. At this point the syringe holder arms **23** are flexed out by the continued force of the second shoulder **24** and allow it to slip through. Now the plunger **9** no longer pushes against the syringe holder **22** but against the stopper **6** for expelling the medicament M from the syringe **3** and injecting it into or through the patient's skin.

(22) When the stopper **6** has nearly bottomed out in the syringe **3** (cf. FIG. 2) the decoupling member **14** has reached a position where it pushes against the latches **12** in a manner to decouple the retraction sleeve **10** from the housing **2**, so the retraction sleeve **10** may slide in distal direction D. Thus the compression spring **8** is no longer grounded with its distal end in the housing **2**. Instead, as soon as the decoupling member **14** has bottomed out at a second abutment **33** the proximal end of the compression spring **8** gets grounded in the housing while the distal end is pulling the retraction sleeve **10** in distal direction D.

(23) Just before the decoupling member **14** decouples the retraction sleeve **10** from the housing **2** the decoupling arms **18** reach an aperture **34** in the retraction sleeve **10** (see FIG. 4) so they are no longer kept from being flexed outward. The decoupling arms **18** are thus pushed outward by the first shoulder **19** pushing against its ramped surfaces so the first shoulder **19** slips through in distal direction as soon as the decoupling member **14** has hit the second abutment **33**.

(24) The syringe holder **22** is taken along in distal direction D by the retraction sleeve **10**, e.g. by a front face **35**. Thus the syringe **3** and needle **4** are retracted into a safe position inside the housing **2**, e.g. into the initial position. The plunger **9**, no longer bearing against the decoupling arms **18** is pulled back too.

(25) In the distal part of the auto-injector **1** a delay box **36** is arranged (see FIG. 5 for details). The delay box **36** comprises a circumferential outer wall **37** with a back collar **38** attached to the housing **2** and a circumferential inner wall **39** with a front collar **40** attached to the retraction sleeve **10**. A volume between the outer wall **37** and inner wall **39** is filled with a viscous fluid, such as silicon grease. As the retraction sleeve **10** is moved in distal direction D the inner wall **39** glides along the outer wall **37** wherein the back collar **38** and front collar **40** increasingly reduce the volume. One or more holes (not shown) provided in a part of the delay box **36** allow the viscous fluid to be pushed out as the volume decreases. The force required to do this slows down the motion of the retraction sleeve **10**.

(26) The retraction sleeve **10** is released by the decoupling member **14** from the housing **2** a certain amount of time or travel before the stopper **6** bottoms out in the syringe **3** and the distal motion of

the retraction sleeve **10** begins. The motion of the retraction sleeve is slowed down by the delay box **36**. Due to a gap **41** between the front face **35** and the syringe holder **22** the retraction sleeve **10** is not yet dragging the syringe back in distal direction D. The plunger **9** is still pushing against the stopper **6** and expelling residual medicament M. As the stopper **6** hits the proximal end of the syringe **3** the stopper **6** and plunger **9** stop while the retraction sleeve **10** is still slowly moving back in distal direction D. The apertures **34** now meet the decoupling arms **18** allowing them to flex out and the plunger **9** to come clear. The retraction sleeve **10** has now travelled back far enough to close the gap **41** so the syringe holder **22**, syringe carrier **7**, syringe **3**, needle **4** and plunger **9** are dragged back in distal direction D.

(27) The cap **25** and the delay box **36** are not restricted to be used with the auto-injector **1** shown in the embodiments. Instead the cap **25** may be combined with any kind of auto-injector with the needle hidden in the housing prior to an injection. The delay box may be combined with any kind of auto-injector for ensuring full delivery of the syringe's contents and reliable triggering of the retraction, irrespective of the spring means or driving means used in the respective auto-injector.

(28) The housing **2** may have at least one viewing window for inspecting the syringe **3**.

(29) The auto-injector **1** may preferably be used for subcutaneous or intra-muscular injection, particularly for delivering one of an analgetic, an anticoagulant, insulin, an insulin derivate, heparin, Lovenox, a vaccine, a growth hormone, a peptide hormone, a proteine, antibodies and complex carbohydrates.

(30) The aforementioned arrangement for coupling the plunger (**9**) to either, the syringe (**3**) or the stopper (**6**), may be applied in any auto-injector having a plunger for forwarding a force of a drive means to a syringe with a stopper. The primary advantage of this arrangement ensures the load from the drive means is not transferred directly to the stopper until the needle is inserted in the patient, thus avoiding a wet injection. The arrangement comprises the syringe holder (**22**) and associated syringe holder arms (**23**), a shoulder (e.g. the second shoulder **24**) on the plunger (**9**), the support of the holder arms (**23**) by an inner surface in order to prevent them from flexing out in a first position and, a widened portion (**2.1**) for allowing them to flex radially and to disconnect from the plunger when in a more proximal position. The spring means or other drive means, the ability to retract the syringe or to forward a needle shroud after injection and other features described herein are not required for the prevention of a wet injection.

LIST OF REFERENCES

(31) **1** auto-injector **2** housing **2.1** widened portion **3** syringe **4** hollow needle **5** protective needle shield **6** stopper **7** syringe carrier **8** spring means, compression spring **8.1** distal end **8.2** proximal end **9** plunger **10** retraction sleeve **11** stop **12** latch **13** end face **14** decoupling member **15** resilient arm **16** stud **17** thrust face **18** decoupling arm **19** first shoulder **20** activating means, trigger button **21** retainer **22** syringe holder **23** syringe holder arm **24** second shoulder **25** cap **26** sheet metal clip **27** barb **28** closing head **29** safety button **30** spring element **31** interlock **32** first abutment **33** second abutment **34** aperture **35** front face **36** delay box **37** outer wall **38** back collar **39** inner wall **40** front collar **41** gap D distal end, distal direction M medicament P proximal end, proximal direction

Claims

1. An auto-injector comprising: a housing; a syringe holder disposed in the housing, the syringe holder configured to retain a syringe comprising a needle such that proximal movement of the syringe relative to the syringe holder is limited, the auto-injector configured such that the syringe holder remains in an initial position relative to the housing as a needle shield is removed from a proximal end of the syringe; a sleeve configured to (i) be coupled to the housing during needle insertion and dose delivery to limit a distal movement of the sleeve relative to the housing, and (ii) be decoupled from the housing during needle retraction to allow the distal movement of the sleeve

relative to the housing; a first member attached to the housing and containing a viscous substance; a second member configured to move relative to the housing and comprising a resilient arm; and a plunger rod comprising a protrusion configured to deflect the resilient arm of the second member, the auto-injector being configured such that prior to needle insertion a proximal movement of the syringe holder relative to the plunger rod is limited by a releasable coupling between the syringe holder and the plunger rod, wherein the auto-injector is configured such that needle retraction is delayed by the viscous substance within the first member.

2. The auto-injector of claim 1, wherein the viscous substance is a grease.

3. The auto-injector of claim 1, wherein the auto-injector is configured such that delaying a distal movement of the syringe holder relative to the housing provides time for substantially all medicament within the syringe to be dispensed from the auto-injector.

4. The auto-injector of claim 1, wherein the auto-injector is configured such that the distal movement of the sleeve relative to the housing is delayed by a deformation of the viscous substance within the first member.

5. The auto-injector of claim 4, wherein the first member has at least one opening configured to allow the viscous substance to be removed from the first member.

6. The auto-injector of claim 1, wherein the plunger rod is releasably held by the sleeve prior to needle insertion and dose delivery.

7. The auto-injector of claim 6, wherein a distal portion of the plunger rod is releasably held by a distal portion of the sleeve prior to needle insertion and dose delivery.

8. The auto-injector of claim 1, comprising a spring disposed in the auto-injector with a distal end engaged to the sleeve and a proximal end engaged to the second member, the spring biasing the sleeve in a distal direction relative to the housing and the second member.

9. The auto-injector of claim 1, comprising a cap removably attached to a proximal end of the housing, wherein the auto-injector is configured to remove the needle shield covering the needle of the syringe when the cap is removed from the proximal end of the housing without moving the needle.

10. The auto-injector of claim 1, wherein the auto-injector is configured to at least partially delay the needle retraction by delaying a start time in which the needle moves distally into the housing after a medicament has been dispensed from the syringe.

11. The auto-injector of claim 10, comprising the syringe containing the medicament.

12. The auto-injector of claim 11, wherein the medicament is a GLP-1 class drug.

13. An auto-injector comprising: a housing; a syringe holder disposed in the housing, the syringe holder configured to retain a syringe comprising a needle such that proximal movement of the syringe relative to the syringe holder is limited, the auto-injector configured such that the syringe holder remains in an initial position relative to the housing as a needle shield is removed from a proximal end of the syringe; a sleeve configured to (i) be coupled to the housing during needle insertion and dose delivery to limit a distal movement of the sleeve relative to the housing, and (ii) be decoupled from the housing during needle retraction to allow the distal movement of the sleeve relative to the housing; a first member disposed in the housing and containing a viscous substance; a second member configured to move relative to the housing and comprising a resilient arm; and a plunger rod comprising a protrusion configured to deflect the resilient arm of the second member, the auto-injector being configured such that prior to needle insertion a proximal movement of the syringe holder relative to the plunger rod is limited by the plunger rod, wherein the auto-injector is configured such that needle retraction is delayed by the viscous substance within the first member.

14. The auto-injector of claim 13, wherein the auto-injector is configured to delay needle retraction to increase an amount of time between dose delivery and needle retraction.

15. The auto-injector of claim 13, wherein the auto-injector is configured to at least partially delay the needle retraction by delaying a start time in which the needle moves distally into the housing after a medicament has been dispensed from the syringe.

16. The auto-injector of claim 15, comprising the syringe containing the medicament.
 17. The auto-injector of claim 13, wherein the viscous substance is a grease.
 18. The auto-injector of claim 13, wherein the plunger rod comprises a shoulder, a distal portion extending distally from the shoulder, and a proximal portion extending proximally from the shoulder.
 19. The auto-injector of claim 18, wherein the auto-injector is configured such that the proximal portion of the plunger rod deflects the resilient arm of the second member while the plunger rod is moving in a proximal direction relative to the housing.
 20. The auto-injector of claim 13, wherein the auto-injector is configured to release the plunger rod from being held by the sleeve by a radial deflection of an additional resilient arm.
 21. An auto-injector comprising: a housing; a syringe holder configured to retain a syringe, the auto-injector configured such that the syringe holder remains in an initial position relative to the housing as a needle shield is removed from a proximal end of the syringe; a sleeve configured to (i) be coupled to the housing during needle insertion and dose delivery to limit a distal movement of the sleeve relative to the housing, and (ii) be decoupled from the housing during needle retraction to allow the distal movement of the sleeve relative to the housing; a first member attached to the housing and containing a viscous substance; a second member comprising a resilient arm; and a plunger rod comprising a protrusion configured to deflect the resilient arm of the second member, the auto-injector being configured such that prior to needle insertion a proximal movement of the syringe holder relative to the plunger rod is limited by the plunger rod, wherein the auto-injector is configured such that needle retraction is delayed by the viscous substance in the first member.
 22. The auto-injector of claim 21, wherein the viscous substance is a grease.
 23. The auto-injector of claim 21, wherein the plunger rod comprises a shoulder, a distal portion extending distally from the shoulder, and a proximal portion extending proximally from the shoulder.
 24. The auto-injector of claim 23, wherein the auto-injector is configured such that the proximal portion of the plunger rod deflects the resilient arm of the second member while the plunger rod is moving in a proximal direction relative to the housing.
 25. The auto-injector of claim 23, wherein the auto-injector is configured to release the distal portion of the plunger rod from being held by the sleeve by a radial deflection of an additional resilient arm.
 26. The auto-injector of claim 25, comprising a cap removably attached to a proximal end of the housing, wherein the auto-injector is configured to remove the needle shield covering a needle of the syringe when the cap is removed from the proximal end of the housing without moving the needle.
 27. The auto-injector of claim 21, wherein the auto-injector is configured to at least partially delay the needle retraction by delaying a start time of the needle retraction.
 28. The auto-injector of claim 21, wherein the plunger rod is configured to engage the syringe holder to hold the syringe holder in the initial position relative to the housing as the needle shield is removed from the proximal end of the syringe such that the syringe remains stationary as the needle shield is removed from the proximal end of the syringe.
 29. The auto-injector of claim 28, comprising the syringe containing a medicament, the auto-injector being configured to dispense the medicament from the syringe prior to needle retraction.
 30. The auto-injector of claim 29, wherein the medicament is a GLP-1 class drug.
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